Alexander DeMarco

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Portfolio of Work: https://ademarco87.github.io

Professional Summary

Systems-focused software engineering graduate with 15+ years of experience in military and emergency services operating in mission-critical environments. Proven ability to translate real-world operational needs into technical solutions, including development of decision-support tools and secure applications. Strong background in systems coordination, resource allocation, and high-pressure problem solving. Seeking systems engineering or IT role supporting operational and secure environments.

Security Clearance: Active Public Trust; previously held DoD Secret Clearance (last active 2018)

Skills

- **Programming & Scripting:** Java, Python, C++, JavaScript, SQL
- **Web & Application Development:** Node.js, Express, REST APIs, Angular, MVC architecture
- **Systems & Infrastructure:** Client-server architecture, application workflows, hardware/software integration, system troubleshooting, basic understanding of network communication (HTTP/HTTPS, request/response)
- **Databases:** MongoDB, MySQL
- **Security:** Authentication (JWT), secure coding practices, encryption concepts, SEI CERT C++ standards
- **Tools & Practices:** Git/GitHub, Agile/Scrum, modular design, debugging and testing

Education

Bachelor of Science in Computer Science: Software Engineering conc. Conferral date: 09/01/2025
Southern New Hampshire University – Manchester, NH

Projects

Secure Client Management System (C++): Refactored a legacy client management system to improve modularity, scalability, and security. Replaced hardcoded credentials with environment-based authentication. Focused on system reliability, input validation, and secure coding practices aligned with SEI CERT standards.

Travlr Getaways Web Application: Built a full stack travel management application using Node.js, Express, and Handlebars, with an Angular SPA for admin functionality. Developed REST APIs, authentication, and dynamic MongoDB data integration.

Animal Shelter Dashboard Application: Developed a secure data dashboard using Python, MongoDB, and Plotly Dash. Implemented authentication, CRUD operations, and real-time visualizations for data-driven decision-making.

Workforce Allocation & Decision Support System (Excel / Web App in Progress): Designed a weighted system to optimize workforce distribution based on tenure, wait time, and hardship factors. Modeled real-world constraints into a scalable decision framework. Currently transitioning to a web-based application using Python and Flask.

Professional Experience

Interim Battalion Chief
Berkeley County EMS | Moncks Corner, SC 02/2026 – Present

- Directed multi-unit operations, ensuring readiness, staffing coordination, and incident response
- Served as decision authority during time-sensitive events, maintaining compliance and continuity
- Bridged field operations and leadership by translating real-world challenges into actionable solutions

Paramedic Lieutenant / FTO 01/2022 – 02/2026
Berkeley County EMS | Moncks Corner, SC

- Supervised personnel and operations in high-pressure environments
- Analyzed operational data to identify trends and improve performance
- Coordinated complex responses, managing resources and strategy during critical incidents
- Trained and evaluated new providers to ensure operational readiness

Tactical Emergency Medical Support (TEMS) Team Leader
Berkeley County EMS | Moncks Corner, SC 01/2024 – 03/2025

- Reestablished and led the county's TEMS program in coordination with law enforcement and command staff following an extended period of inactivity.
- Authored TEMS policies, procedures, and operational guidelines from the ground up, defining command structure, deployment criteria, and medical oversight.
- Provided command-and-control medical leadership during tactical operations, ensuring integration with incident command and alignment with law enforcement objectives.
- Advised and collaborated with other TEMS teams on program development and best practices and developed training standards that enabled sustained operational capability; the program remains active today.

Paramedic Crew Chief / EMT 03/2018 - 01/2022
Berkeley County EMS | Moncks Corner, SC

- Delivered advanced 911 emergency care by assessing injury and illness severity, implementing treatment protocols per clinical guidelines, and coordinating care with on-scene medical personnel to ensure optimal patient outcomes.
- Ensured timely and accurate completion of electronic patient records, meticulously documenting clinical interventions, patient assessments, and treatment plans to maintain compliance with regulatory standards.
- Conducted routine operational duties, including equipment maintenance, inventory checks, and facility sanitation, to ensure readiness of medical apparatus and work environments for emergency response.

Senior Healthcare Specialist 01/2010 – 12/2018
United States Army | Fort Hood, Texas

- Delivered medical support in high-risk operational environments that require strict adherence to protocol and documentation.
- Coordinated field operations and evacuations in alignment with command objectives.
- Maintained accountability for $100K+ in mission-critical equipment and trained personnel on standardized procedures.

Other Education

Associate of Science in Healthcare Management July 2016
Kaplan University (now Purdue Global) – Online